

Algorithm

Blast Design Parameters

Step 1: Input the Blasthole Diameter, Linear Mass Density, explosive density, burden-spacing ratio, bench height, technical powder factor, stemming length, subdrill, burden, spacing, actual powder factor as variables; BD, M, d, rA, BH, PFtech, SL, SD, B, S, PFact respectively.

Step 2: Use the formula $M = (d * BD^2) / 1273$ to calculate the linear mass density.

Step 3: Check if the blast has occurring close to private property by checking if the distance to surroundings is greater than 1000 meters.

Step 4: If the distance is greater than 1000 meters then calculate the stemming length $SL = 30 * BD$.

Step 5: If the distance is less than 1000 meters then calculate the stemming length using $SL = 20 * BD$.

Step 6: Display the Stemming length value.

Step 7: Calculate the burden by using the formula $B = \sqrt{\frac{M * (BH - SL)}{(rA * PFtech * BH)}}$

Step 8: Display the burden value.

Step 9: Calculate the subdrill value by using the formula $SD = \frac{1}{4} * B$

Step 10: Display the subdrill value.

Step 11: Check If the rock type is Hard rock or soft rock from the input.

Step 12: If the rock type is Hard rock then calculate the actual powder factor using the formula $PFact = \frac{M * (BH - SL + SD)}{B * S * BH}$

Step 13: If the rock type is Soft rock then calculate the actual powder factor using the formula $PFact = \frac{M * (BH - SL - SD)}{B * S * BH}$

Step 14: End.

Fragmentation Calculator

Step 1: Input parameters the Rock factor, Relative Explosive Energy, PFtech, density of explosive, K constant, blasthole Diameter, Bench height, mass of explosive per hole, mass density, Mean Fragmentation size and stemming length as variables; A, REE, PFtech, p, K, D, BH, Q, M, FragmentationSize and SL

Step 2: Calculate $M = (d * BD^2) / 1273$

Step 3: Calculate the mass of explosive per hole: $Q = M * (BH - SL)$

Step 4: Calculate FragmentationSize = $A * K^{-0.8} * Q^{0.167} * \frac{115^{0.633}}{REE}$

Step 5: Display FragmentationSize